

Prompt: Drawing from the resources shared guide me on how I will do exercise 5 don't explicitly state the answer, guide me to come up with the code for me,DON'T EXPLICITLY STATE IT."""

Exercise 5: Pandas Data Analysis

Analyze a dataset of student performance.

"""

```
import pandas as pd
```

```
import numpy as np
```

```
# Create sample dataset
```

```
np.random.seed(42)
```

```
n_students = 200
```

```
data = {
```

```
    'student_id': range(1000, 1000 + n_students),
```

```
    'major': np.random.choice(['CS', 'Math
```

```
        ', 'Physics', 'Biology'], n_students),
```

```
    'year': np.random.choice([1, 2, 3, 4], n_students),
```

```
    'exam_score': np.random.normal(75, 10, n_students).clip(0, 100),
```

```
    'assignments_completed': np.random.randint(0, 11, n_students),
```

```
    'hours_studied': np.random.normal(15, 5, n_students).clip(1, 40)
```

```
}
```

```
df = pd.DataFrame(data)
```

```
# Introduce some NaN values
```

```
df.loc[np.random.choice(n_students, 10), 'exam_score'] = np.nan
```

```
df.loc[np.random.choice(n_students, 5), 'hours_studied'] = np.nan
```

```
# Task 1: Data Cleaning and Exploration (10 points)
```

```
# TODO: Display basic information about the dataset  
# TODO: Identify and count missing values  
# TODO: Fill missing exam_score with the mean score for the student's major  
# TODO: Fill missing hours_studied with the median for the student's year
```

RESULT:

Task 1: Data Cleaning and Exploration (10 points)

Step 1: Write a code to display Basic Information

2. Write the code to count missing values

3. The task says:

Fill missing exam_score with the mean score for the student's major

First calculate the mean exam score for each major.

Then fill missing values.

4. The task says:

Fill missing hours_studied with the median for the student's year

5. Verify missing values and print it by checking if it is null